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AL AKHAWAYN UNIVERSITY

School of Science and Engineering

Inspire, Impact & Transform

AneuXplain:

An Explainable AI System for Intracranial Aneurysm Rupture Risk Prediction

Capstone Design Report

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Supervised by:

Dr. Oualid Kafi

Al Akhawayn University in Ifrane — Spring 2026

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The supervisor and the student (the Capstoner) agree that:

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2. The results described in this report have not previously been published.

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Student Statement

I, Mohamed Al Mamoun Tantaoui El Araki, hereby declare that this capstone report and the work it describes are the result of my own efforts carried out under the supervision of Dr. Oualid Kafi at the School of Science and Engineering, Al Akhawayn University in Ifrane. All sources of information and third-party datasets have been acknowledged and cited. The evaluation protocols were designed to be transparent, reproducible, and honest about both the strengths and the limitations of the system.

Approved by the Supervisor,

Dr. Oualid Kafi

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I would like to express my sincere gratitude to Dr. Oualid Kafi for his guidance throughout this capstone project. His direction — particularly his insistence on the rigorous manual validation of 50 individual subject prototypes on real clinical data — shaped the scientific character of this work and pushed it beyond a simple proof of concept.

I also acknowledge the authors of the two public datasets used in this project: Yan et al. for IntrA (CVPR 2020), which made this project possible by providing segmented 3D artery meshes, and Di Noto et al. for the Lausanne TOF-MRA Aneurysm Cohort (Neuroinformatics 2022, OpenNeuro ds003949), which enabled cross-dataset validation on real clinical volumes.

Finally, I acknowledge the School of Science and Engineering at Al Akhawayn University for the academic environment that made this project possible.

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Abstract (English)

Intracranial aneurysms are localized weak spots in the blood vessels of the brain. When one ruptures, the result is a subarachnoid hemorrhage with high mortality. Deciding whether to treat an unruptured aneurysm remains a delicate clinical judgment that balances rupture risk against treatment risk. A tool that helps a clinician understand *why* a specific aneurysm is high-risk — not just whether it is — would therefore be more clinically useful than a black-box classifier.

This capstone presents AneuXplain, an explainable artificial intelligence system for intracranial aneurysm rupture risk prediction. The system combines three independent analytical pathways: (i) a PointNet-based deep learning risk predictor that scores 3D artery meshes, (ii) a gradient-based spatial heatmap module that highlights which mesh regions drive the prediction, and (iii) a morphological analysis module that extracts eight clinically established geometric parameters and presents them in a blood-test-style report with literature-based reference thresholds.

The risk predictor was trained on the IntrA dataset (2012 point-cloud samples from 103 patients) using a patient-level 70/15/15 train/validation/test split that prevents the data leakage typical of case-level random splits. On the held-out test set of 16 patients (312 samples), the model achieves a ROC-AUC of 0.834 and an accuracy of 84.9% at threshold 0.5, with a balanced operating point at threshold 0.27 giving recall 76.9% and specificity 76.2%.

The AneuXplain system was manually validated on 50 subjects from the Lausanne TOF-MRA Aneurysm Cohort (ds003949), using a supervised verification protocol to ensure the clinical plausibility of each generated report. The cohort comprised 25 aneurysm patients and 25 healthy controls; 48 of 50 subjects completed the full pipeline end-to-end (96%) with a mean system response time of 13.9 seconds per subject per manual invocation. Cross-dataset ROC-AUC on Lausanne was 0.557, substantially lower than the within-dataset value, with every subject flagged as out-of-distribution by the harmonization module (mean edge-length z-score ≈ -116). This distribution-shift finding, rather than being hidden, is treated as a principal scientific outcome of the work: it motivates the system's explainability design and highlights a real challenge in translating medical AI across scanner, protocol, and preprocessing differences.

Keywords: intracranial aneurysm, explainable AI, PointNet, TOF-MRA, cross-dataset evaluation, medical imaging.

Résumé (Français)

Les anévrismes intracrâniens sont des fragilités localisées des vaisseaux sanguins du cerveau. Leur rupture provoque une hémorragie sous-arachnoïdienne à forte mortalité. Décider de traiter un anévrisme non rompu reste un jugement clinique délicat. Un outil qui aide le clinicien à comprendre *pourquoi* un anévrisme donné est à haut risque serait donc plus utile qu'un classifieur boîte noire.

Ce projet de fin d'études présente AneuXplain, un système d'IA explicable pour la prédiction du risque de rupture d'anévrismes intracrâniens, combinant un prédicteur PointNet, une carte de chaleur spatiale par gradient, et un module d'analyse morphologique à huit paramètres. Le prédicteur a été entraîné sur IntrA (2012 échantillons, 103 patients) avec un partitionnement au niveau patient, atteignant une AUC de 0,834 sur un ensemble de test de 16 patients. Le système AneuXplain a ensuite été validé manuellement sur 50 sujets de la cohorte Lausanne TOF-MRA (ds003949) à l'aide d'un protocole de vérification supervisé, sujet par sujet ; 48 sur 50 sujets ont été traités de bout en bout. L'AUC inter-ensembles était de 0,557 avec une dérive de distribution confirmée (tous les sujets hors distribution selon l'harmonisation). Ce décalage est traité comme un résultat principal du travail.

Mots clés : anévrisme intracrânien, IA explicable, PointNet, TOF-MRA, évaluation inter-ensembles, imagerie médicale.

Chapter 1 : Introduction

An intracranial aneurysm is a localized bulge in the wall of a brain artery. Most aneurysms go undetected and never cause symptoms, but when one ruptures, the blood that escapes into the space around the brain causes a subarachnoid hemorrhage. The reported 30-day mortality of aneurysmal subarachnoid hemorrhage is approximately 35–40%, and a significant fraction of survivors are left with permanent neurological deficits. For this reason, aneurysms that are discovered incidentally on imaging pose a difficult clinical question: treat now and accept the non-trivial risk of endovascular or surgical intervention, or observe and accept the risk that the aneurysm ruptures before the next scan.

The decision is guided by aneurysm size, location, shape, and patient factors. Clinical risk scores such as PHASES (Greving et al., 2014) structure this decision, but they reduce complex 3D geometry to a handful of numbers and a single size threshold. Recent research — ISUIA, Ujiie et al. — has shown that detailed shape features carry independent predictive value beyond size alone.

Deep learning offers a way to analyze 3D geometry directly. The difficulty is that a deep learning model is, by default, a black box. A clinician asked to trust a score of 0.83 out of 1.00 needs to know what drove that score. If the model has been influenced by a spurious correlation — a specific scanner, a specific patient cohort, or an artifact of image pre-processing — then its score is dangerous, not useful. This is especially true in medical imaging, where the distribution of inputs can shift between training and deployment in ways that are invisible to the end user.

AneuXplain is designed around this observation. Instead of a single score, the system produces three complementary outputs: a risk score from a deep learning model, a spatial heatmap that shows which part of the aneurysm drove the score, and a structured morphological report with eight measurements that a clinician can cross-check against published thresholds. The three outputs can reinforce each other — or disagree — and it is that interaction, not any one of them in isolation, that is intended to be clinically useful.

Research question. The scientific question pursued by this capstone can therefore be stated as follows: *Can a multi-pathway explainable AI system for intracranial aneurysm rupture risk prediction be constructed end-to-end on real clinical inputs, and do its outputs remain informative when the underlying deep learning classifier is exposed to cross-dataset distribution shift?* This question has two coupled components — a system-design component (whether such a multi-pathway pipeline can be built and operated reliably) and an empirical component (how its three pathways behave when the input distribution differs from the training distribution). Chapter 7 addresses the first component through a rigorous internal evaluation on the

IntrA dataset, and Chapters 7 and 8 address the second through a supervised, subject-by-subject manual validation on the Lausanne TOF-MRA Aneurysm Cohort.

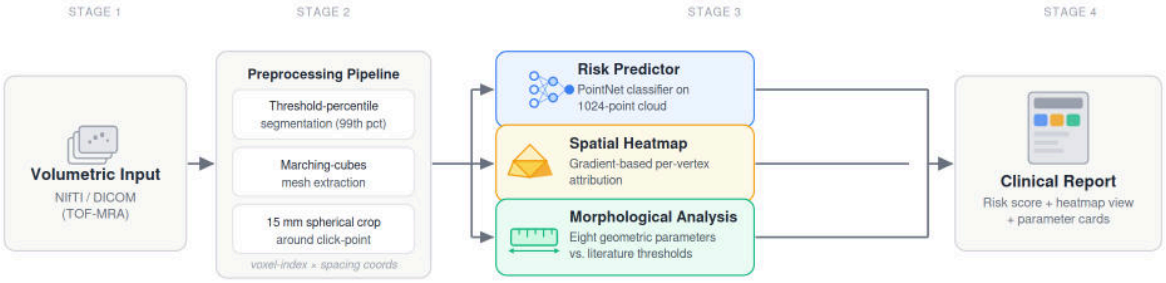


Figure 1.1: Three-module architecture of the AneuXplain system. A volumetric TOF-MRA input is preprocessed (segmentation, mesh extraction, spherical crop) before entering three parallel analysis pathways — Risk Predictor (blue), Spatial Heatmap (orange), and Morphological Analysis (green) — whose outputs are consolidated into a structured clinical report.

Chapter 2 : Problem Statement

The core problem this capstone addresses is the following. Black-box deep learning classifiers for aneurysm risk, although they sometimes achieve impressive reported accuracy on in-dataset benchmarks, are not directly usable in a clinical setting for three reasons.

First, they do not explain why. A score, on its own, gives a clinician no way to audit the prediction against their anatomical knowledge. This is both a trust problem and a safety problem.

Second, they are fragile across datasets. A model that achieves 0.90 AUC on the dataset it was trained on often collapses to 0.60–0.75 AUC when applied to a second dataset from a different hospital, different scanner, or different pre-processing pipeline. This capstone's own cross-dataset evaluation reproduces the phenomenon on real data.

Third, the clinical question is not simply "is this an aneurysm?" It is "will this specific aneurysm rupture, and why?" The second question cannot be answered by any classifier alone — it requires connecting predictions to anatomy and to published risk factors.

AneuXplain therefore does not aim to be the most accurate standalone classifier on a single benchmark. It aims to be an honest, auditable, decision-support tool in which the risk score, the spatial heatmap, and the morphological report are exposed together to the clinician. The scientific question of this capstone is whether such a multi-pathway explainable system can be constructed end-to-end on real clinical inputs — and whether its outputs remain informative even when the underlying classifier shows known limitations under distribution shift.

Chapter 3 : Project Objectives

The capstone pursues five concrete objectives:

1. **Objective 1.** Design and implement an end-to-end pipeline that ingests a brain angiography volume (NIfTI or DICOM), segments the cerebral vessels, extracts a 3D mesh, and produces a risk score, a spatial heatmap, and a morphological report.

2. **Objective 2.** Train a PointNet-based risk predictor on the IntrA dataset using a rigorous patient-level split to prevent data leakage, and report its performance honestly on a held-out test set.

3. **Objective 3.** Implement a morphological analysis module that computes eight clinically established parameters (neck width, dome height, maximum dome diameter, aspect ratio, dome-to-neck ratio, irregularity index, volume, surface area) and presents them with literature-based reference ranges.

4. **Objective 4.** Validate the full pipeline on 50 subjects from an independent public dataset (Lausanne TOF-MRA Aneurysm Cohort, ds003949) and measure cross-dataset generalization, pipeline reliability, runtime, and distribution shift.

5. **Objective 5.** Produce a web-based interface (FastAPI backend, React + Three.js frontend) that makes the full system usable and inspectable by a non-expert user.

A deliberate non-objective is worth stating. Early in the project, a healing-pipeline module was explored that would take a pathological aneurysm mesh and generate a hypothetical "healed" version. After consulting with the supervisor, this module was removed. No treatment actually restores a vessel to its pre-aneurysm shape, and a generated healed mesh would therefore be clinically misleading. The clinical value of AneuXplain lies in explaining why an aneurysm is dangerous, not in imagining a counterfactual that could never be achieved.

Chapter 4 : Literature Review

4.1 Clinical scoring of aneurysm rupture risk

The clinical literature on aneurysm risk prediction rests on several landmark studies. The International Study of Unruptured Intracranial Aneurysms (ISUIA; Wiebers et al., 2003) established size as the dominant predictor of rupture and showed that small aneurysms rarely rupture over a 5-year horizon. Ujiie et al. (2001) introduced the aspect ratio — dome height divided by neck width — as a strong size-independent predictor. The PHASES score (Greving et al., 2014) integrates population, hypertension, age, size, earlier hemorrhage, and location into a single five-year rupture risk estimate. Backes et al. (2015) later introduced the ELAPSS score for aneurysm growth. These clinical scores, although useful, are coarse summaries of a complex 3D shape.

4.2 Morphological analysis from 3D geometry

More detailed morphological analysis uses 3D mesh or surface representations of the aneurysm to compute geometry beyond size. Parameters that appear repeatedly in the literature include neck width, dome height, maximum dome diameter, aspect ratio, dome-to-neck ratio, non-sphericity / irregularity index, volume, and surface area. Several studies (Dhar et al., 2008; Xiang et al., 2011) have shown that combinations of these parameters outperform size alone for distinguishing ruptured from unruptured aneurysms.

4.3 Deep learning on 3D vascular meshes

Deep learning approaches to 3D vascular geometry have followed several tracks. PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b) are the dominant architectures for learning directly from point clouds, and they have been applied to aneurysm classification by several groups, with reported AUCs between 0.80 and 0.90 on the IntrA dataset.

Recent work has extended this baseline along three lines that are directly relevant to AneuXplain. First, several groups have refined PointNet-style classifiers on aneurysm data. Cao et al. (2024) trained a PointNet++ classifier on 623 aneurysms drawn from the AneuX morphology database and reported a within-dataset cross-validation AUC of 0.85, but observed a drop to 0.71 on an external dataset — explicitly framed as a generalizability concern of the same kind that the present capstone investigates. Shou et al. (2024) integrated PointNet-extracted hemodynamic-cloud features with classical machine learning classifiers to discriminate ruptured from unruptured aneurysms on a CFD-augmented cohort. Most recently, Wang et al. (2026), in their M-PointNet paper, proposed a multi-layer embedded architecture built on

PointNet++ that reports 91.96% accuracy and an F1-score of 0.923 on intracranial aneurysm classification on the IntrA dataset, illustrating that the basic point-cloud backbone continues to receive active architectural attention.

A second line of recent work focuses on *generative* models of vascular geometry rather than classification. Ding et al. (2025), in their MICCAI 2025 paper introducing AneuG, propose a two-stage variational autoencoder that generates intracranial aneurysm meshes conditioned on clinically meaningful morphological markers (aspect ratio, neck width, lobulation index), demonstrated on 116 middle-cerebral-artery-bifurcation aneurysms from the AneuX database. Chen et al. (2025), in their MCMeshGAN paper, propose a conditional mesh-to-mesh generative adversarial network that predicts longitudinal aneurysm growth conditioned on patient age, sex, and a target time interval, evaluated on 590 longitudinal records of thoracic aortic aneurysms. These generative systems target shape synthesis and growth prediction rather than rupture-risk classification, but they share with the present work an emphasis on mesh-level representations and on conditioning the model on clinically interpretable parameters.

A third line of work is the systematic study of distribution shift itself. Yoon et al. (2024), in a comprehensive survey of domain generalization for medical image analysis published in Proceedings of the IEEE, document that performance degradation of 10–25% on out-of-distribution data is the norm rather than the exception, and they catalog data-, feature-, model-, and analysis-level techniques for mitigating it. The cross-dataset finding reported in Chapter 7 of this capstone — a drop from 0.834 to 0.557 ROC-AUC between IntrA and the Lausanne cohort — sits firmly inside the regime that this body of work describes.

Two methodological issues recur across all three lines. First, many reported performance numbers are obtained with case-level random splits, which allow fragments of the same patient to appear in both training and test sets. Second, cross-dataset evaluation is rare: most published numbers reflect within-dataset performance only. Both issues are directly addressed in this capstone.

To position AneuXplain explicitly within this literature, Table 4.1 contrasts the present work with the most relevant published systems for 3D vascular geometry along three axes that matter most for clinical translation: whether external (cross-dataset) evaluation is performed, what form of explainability the system exposes to the user, and whether the system delivers an end-to-end interactive interface alongside the model. Existing systems tend to specialize in one axis — a strong classifier, a generative shape model, or a mesh-synthesis architecture — whereas AneuXplain is deliberately designed as a combination: a classifier with bounded expectations, paired with structured morphological evidence and gradient-based localization that remain interpretable even when the classifier itself is uncertain.

Table 4.1: Positioning of AneuXplain relative to representative prior work on 3D vascular geometry. Cells marked "not reported" indicate that the published description of the system does not, to the author's knowledge, include a cross-dataset evaluation; this is the gap that motivates AneuXplain's external validation on the Lausanne TOF-MRA cohort.

System	Primary task	Cross-dataset evaluation	Explainability	Interactive interface
PointNet (Qi et al., 2017a)	General 3D classification	N/A (baseline architecture)	None	No
PointNet++ classifier (Cao et al., 2024)	Aneurysm rupture-risk classification	Yes (HUG → @neurIST: AUC 0.85 → 0.71)	Score only	No
M-PointNet (Wang et al., 2026)	Aneurysm classification + segmentation	Not reported	Score + segmentation mask	No
MCMeshGAN (Chen et al., 2025)	Longitudinal aneurysm growth prediction	Not reported	None (generative)	No
AneuG (Ding et al., 2025)	Conditional aneurysm shape synthesis	Not reported (in-dataset only)	None (generative)	No
AneuXplain (this work)	**Risk prediction + morphology + spatial localization**	**Yes (IntrA → Lausanne, 50 subjects; AUC 0.834 → 0.557)**	**Risk score + gradient heatmap + 8-parameter morphological report**	**Yes (FastAPI + React/Three.js)**

The novelty claim of this capstone is therefore not architectural — the underlying PointNet backbone is well-established — but methodological and integrative: a transparent patient-level training protocol, a manually verified cross-dataset evaluation that is honest about distribution shift, and a multi-pathway explainable interface that surfaces the complete chain of evidence to a non-expert user.

4.4 The IntrA dataset and the Lausanne cohort

The IntrA dataset (Yan et al., 2020) provides 2012 3D mesh samples from 103 patients, organized into generated vessel segments (1694 healthy), generated aneurysm segments (215 pathological), and complete artery trees (103). The dataset has been widely used for supervised learning of aneurysm classifiers.

The Lausanne TOF-MRA Aneurysm Cohort (Di Noto et al., 2022) is a larger public dataset of raw TOF-MRA volumes: 284 subjects in total, with 157 patients having at least one unruptured aneurysm and 127 healthy controls. An important methodological feature of this dataset is the

use of weak spherical labels: instead of drawing the precise aneurysm outline voxel-by-voxel, the radiologist placed a sphere of appropriate size that fully enclosed the aneurysm. This weak-label scheme is four times faster to generate than voxel-wise annotation but is an upper bound on the true aneurysm extent, which is relevant for interpreting this capstone's cross-dataset results.

4.5 Explainability in medical AI

Explainable AI (XAI) in medical imaging is a rapidly maturing subfield rather than a settled one. Early work converged on a set of post-hoc techniques: saliency maps and gradient-based heatmaps (Selvaraju et al., 2017), example-based explanations, concept activation vectors, and human-interpretable summaries. Recent surveys synthesize this work and document where the field is moving. Chaddad et al. (2023), in a broad survey of XAI techniques in healthcare, catalogue the principal post-hoc methods (Grad-CAM, LIME, SHAP, attention visualizations) used across radiology, pathology, and dermatology and emphasize that no single explanation type has been shown to be universally faithful to a model's underlying decision process. Bhati et al. (2024), focusing specifically on visualization techniques for deep models in medical imaging, reach a similar conclusion and explicitly recommend combining complementary explanation pathways rather than relying on a single saliency map. Both surveys identify the same gap that AneuXplain is designed to address: a single attribution map, however sophisticated, is hard to audit against clinical knowledge in isolation.

For 3D mesh-based models, gradient attribution with respect to vertex positions produces a per-vertex scalar field that can be visualized as a heatmap on the mesh surface — the technique used for AneuXplain's heatmap module. Clinical translation generally favors multi-pathway explanations — a score plus a localization plus a structured measurement summary — over any single explanation type in isolation, both because no single method has been shown to be universally faithful (Chaddad et al., 2023; Bhati et al., 2024) and because clinicians audit predictions against multiple sources of evidence simultaneously. The design of AneuXplain follows this principle.

Chapter 5 : Methodology

5.1 Research Design

This work is quantitative in character. It consists of (i) designing and implementing an explainable AI pipeline, (ii) training a deep learning model on a public dataset under a rigorous evaluation protocol, and (iii) validating the full pipeline on a separate public dataset not used during training. No human participants were recruited; both datasets used are publicly available and de-identified at the source. The evaluation is therefore a retrospective secondary analysis of open data.

The research design emphasizes three principles: reproducibility (every experimental artifact is produced by a committed script, and random seeds are fixed), separation of training and evaluation cohorts (IntrA for training with patient-level split, Lausanne for external validation), and honest reporting (both positive and negative findings are included, with explicit discussion of distribution shift and failure cases).

5.2 Population and Data Sources

5.2.1 IntrA dataset (training and internal evaluation)

The IntrA dataset (Yan et al., 2020) was used for training and internal evaluation of the risk predictor. The dataset contains 2012 3D point-cloud samples extracted from 103 unique patients. For training, the three categories were used as follows: generated vessel segments (1694 samples, label = 0 / healthy), generated aneurysm segments (215 samples, label = 1 / diseased), and complete artery trees (103 samples, label = 1 / diseased). The annotated subfolder of IntrA was not used for supervised training.

A patient-level 70/15/15 split was applied. Because IntrA filenames follow the convention `ArteryObjAN<patient>-<fragment>.obj`, patient identifiers were extracted with a regular expression. Whole patients were then randomly assigned to training, validation, or test using a fixed seed of 42, producing 72 training patients (1417 samples), 15 validation patients (283 samples), and 16 test patients (312 samples). This split is persisted in `data/patient_splits.json` and is used unchanged across all experiments. A verification step explicitly confirmed that no patient identifier appears in more than one split.

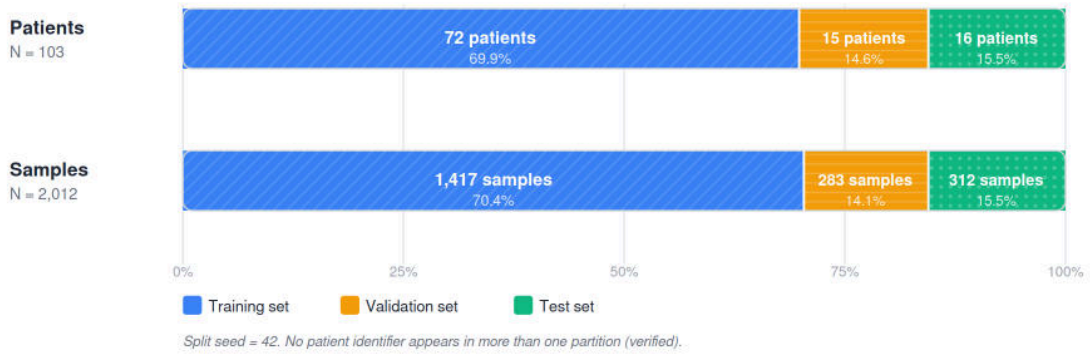


Figure 5.1: Patient-level train/validation/test split of the IntrA dataset. Proportions are consistent across patients (top bar) and samples (bottom bar), confirming that the split boundaries were drawn at the patient level to prevent data leakage. Split seed = 42; no patient identifier appears in more than one partition (verified).

5.2.2 Lausanne TOF-MRA Aneurysm Cohort (cross-dataset validation)

For the prototype evaluation, 50 subjects were drawn from OpenNeuro ds003949 (Di Noto et al., 2022). A balanced sub-cohort was selected with 25 patients (with at least one aneurysm) and 25 healthy controls. Selection favored acquisition-date diversity to avoid over-representing sessions acquired near each other in time.

For the 25 patients, the expert-annotated lesion masks provided in the dataset (`derivatives/manual_masks/s` were downloaded. For two patients (sub-121 and sub-126), the dataset contained two aneurysm masks per subject; the lesion with the larger non-zero voxel count was selected for each, and the decision logged. For controls, only brain masks are available (since no aneurysm is present), and the aneurysm click-point strategy is described in section 5.3.

5.3 Data Collection Methods

5.3.1 System architecture

AneuXplain is implemented as a FastAPI backend (Python, PyTorch (Paszke et al., 2019), SciPy, Trimesh, SimpleITK (Yaniv et al., 2018), scikit-image (van der Walt et al., 2014)) paired with a React 19 frontend using Three.js for 3D visualization. Model training was conducted on Google Colab (free tier, Tesla T4 GPU). All training, evaluation, and validation code is

versioned on GitHub.

The backend exposes two principal analytical entry points. First, the `/analyze` endpoint accepts a pre-extracted mesh file (`.obj`, `.ply`, `.stl`, `.off`) and returns the full three-pathway analysis. Second, a set of `/dicom` endpoints accepts raw volumetric data and handles the upstream conversion: `/dicom/upload` parses and stores the volume, `/dicom/segment` applies threshold-percentile segmentation and marching-cubes mesh extraction, and `/dicom/crop-and-analyze` crops a spherical region around a clinician-supplied click-point before running the analysis.

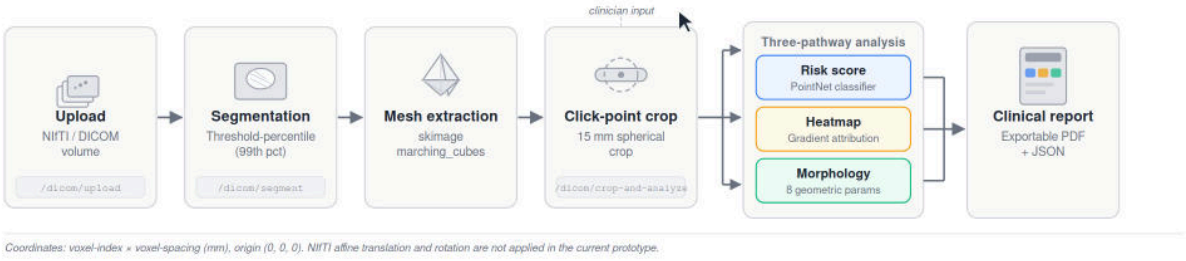


Figure 5.2: End-to-end pipeline for processing a volumetric input. Raw NIfTI/DICOM volumes are segmented, converted to a 3-D mesh, and cropped around a clinician-selected click-point. The cropped region enters three parallel analysis pathways before being consolidated into an exportable clinical report. Coordinates are in voxel-index $\times$ voxel-spacing (mm), origin $(0, 0, 0)$; the NIfTI affine translation and rotation are not applied in the current prototype.

5.3.2 Risk predictor module

The risk predictor is a PointNet-based binary classifier operating on 1024-point clouds sampled from the input mesh. The architecture follows the standard PointNet design: per-point MLPs followed by symmetric max-pooling aggregation, followed by fully connected layers with batch normalization and dropout. The output is a single logit, trained with `BCEWithLogitsLoss`. A `pos_weight` of 5.03 (ratio of healthy to diseased training samples) is applied to compensate for class imbalance; `WeightedRandomSampler` was initially combined with `pos_weight` but was removed in the final run after empirical evidence that the double class correction caused probability compression and unstable validation accuracy.

Training uses Adam (Kingma & Ba, 2015) with learning rate 2×10^{-4}, weight decay 1×10^{-4}, a cosine annealing schedule over 80 epochs, batch size 16, and early stopping with patience 25 on the prediction-gap metric. Five checkpoints are preserved at the end of training (best validation loss, best validation AUC, best validation accuracy, best gap, last epoch), and the best-validation-AUC checkpoint is deployed as the canonical model. Training augmentation includes uniform random rotation around the Z-axis, per-coordinate Gaussian jitter (standard

deviation 0.01), and uniform random scaling (0.9–1.1); augmentation is applied only to the training subset and not to validation or test.

5.3.3 Spatial heatmap module

The spatial heatmap is computed by gradient-based attribution. For an input mesh, the PointNet output is differentiated with respect to the vertex coordinates, and the resulting per-vertex gradient magnitude is normalized using the 95th percentile to suppress outliers. The values are then mapped through an HSL color interpolation, producing a per-vertex scalar field displayed as a heatmap on the mesh surface. The heatmap is rendered in the Three.js frontend with a shared camera pose between the mesh view and the heatmap view, so that the user can compare them directly.

Formally, given the trained classifier f_θ and an input point cloud $X = \{x_1, \ldots, x_n\}$ with $x_i \in \mathbb{R}^3$, the per-vertex attribution is computed as the gradient magnitude

$$g_i = \|\nabla_{x_i} f_\theta(X)\|_2 \,, \tag{5.1}$$

which is then normalized using the 95th percentile of $\{g_i\}$ to bound the displayed range and suppress outlier vertices,

$$\tilde{g}_i = \mathrm{clip}\left(\frac{g_i}{Q_{95}(\{g_j\}_{j=1}^n)},\ 0,\ 1 \right). \tag{5.2}$$

The normalized values $\tilde{g}_i \in [0,1]$ are then mapped through an HSL interpolation (blue for low attribution, red for high) and rendered on the mesh surface.

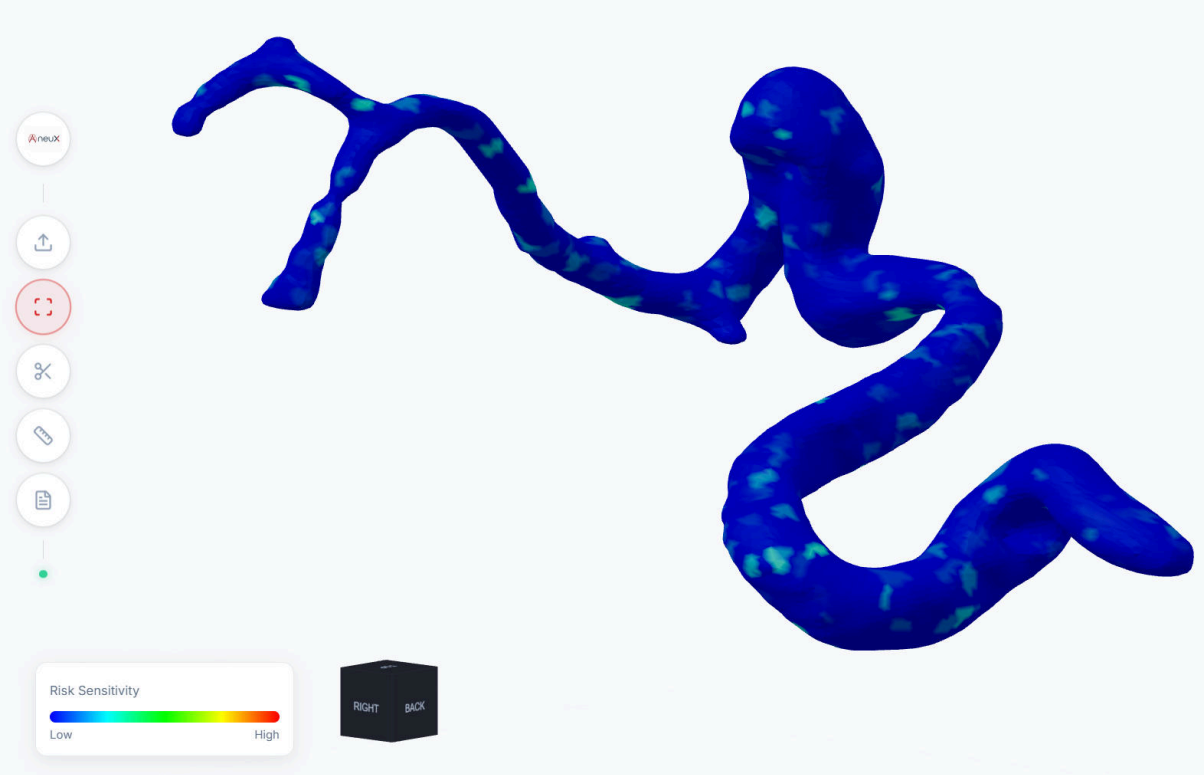


Figure 5.3: Example of gradient-based spatial heatmap on a mesh. Left: original mesh rendered in the AneuXplain Three.js frontend. Right: same mesh with per-vertex gradient attribution displayed as a heatmap, using 95th-percentile normalization and HSL color interpolation (blue low, red high). Both panels share the same camera pose, enabling direct visual comparison.

5.3.4 Morphological analysis module

The morphological analysis module extracts eight geometric parameters from the cropped aneurysm mesh and presents each one in a blood-test-style card showing the measured value, the literature-based reference range, and a colour-coded risk level. This subsection defines each parameter formally so that the implementation in the project repository is unambiguous and reproducible.

Let the mesh be a pair $\mathscr{M} = (\mathscr{V}, \mathscr{T})$, where $\mathscr{V} = \{v_1, \ldots, v_n\} \subset \mathbb{R}^3$ is the set of vertices and $\mathscr{T} = \{(v_a^i, v_b^i, v_c^i)\}_{i=1}^N$ is the set of triangular faces. Two derived subsets are also used below: $\mathscr{V}_{\text{neck}}$, the vertices lying on the neck cross-section identified by the cascade described later in this subsection, and $\mathscr{V}_{\text{dome}}$, the vertices lying on the aneurysm dome (the side of the neck plane opposite to the parent vessel).

Surface area and volume. Surface area is the sum of the areas of all triangular faces, computed via the cross-product formula:

$$S = \sum_{i=1}^{N} \tfrac{1}{2} \left\| (v_b^i - v_a^i) \times (v_c^i - v_a^i) \right\|. \tag{5.3}$$

Volume is computed via the divergence theorem as the magnitude of the sum of signed tetrahedral volumes formed by each face and the origin (assumes a closed, consistently-oriented mesh; the cropping procedure ensures this in practice):

$$V = \frac{1}{6}\left|\sum_{i=1}^{N} v_a^i \cdot \left(v_b^i \times v_c^i\right)\right|. \tag{5.4}$$

Neck detection cascade. The neck is the cross-section connecting the parent vessel to the aneurysm dome and is the most clinically delicate parameter to extract. AneuXplain uses a three-tier cascade:

1. **Centerline-based (primary).** A 1D centerline curve $C(t)$, $t \in [0, L]$, is extracted from $\mathscr{M}$ via a Voronoi-skeletonization-based approach. The neck is identified as the local minimum of cross-sectional diameter $d(t)$ within the parent-vessel-to-dome transition window $[t_a, t_b]$:

$$t_{\text{neck}} = \arg\min_{t \in [t_a, t_b]} d(t). \tag{5.5}$$

2. **Curvature-based (fallback 1).** If centerline extraction fails (typically due to degenerate topology), the neck ring is identified as the closed contour of vertices where the discrete mean curvature $H(v)$ changes sign across the parent-vessel-to-dome transition.

3. **Boundary-based (fallback 2).** If both methods above fail, the neck is identified from the boundary edges of the cropped mesh. This method is used only as a last resort because boundary edges produced by spherical cropping can numerically mimic an anatomical neck and produce false readings; such cases are flagged in the report.

Neck width. Once $\mathscr{V}_{\text{neck}}$ is identified with neck-plane centroid c_n, the neck width is defined as twice the maximum distance from the centroid to the neck-plane vertices:

$$W_n = 2 \cdot \max_{v \in \mathscr{V}_{\text{neck}}} \|v - c_n\|_2. \tag{5.6}$$

Dome height. With the neck plane identified by its unit normal $\hat{n}$ and centroid c_n, dome height is the maximum signed projection of dome vertices onto $\hat{n}$:

$$H_d = \max_{v \in \mathscr{V}_{\text{dome}}} \hat{n} \cdot (v - c_n). \tag{5.7}$$

Maximum dome diameter. The maximum dome diameter is the largest pairwise distance between dome vertices, projected onto the plane orthogonal to the dome axis $\hat{n}$:

$$D_{\max} = \max_{v_i, v_j \in \mathcal{V}_{\text{dome}}} \left\| \Pi_{\hat{n}}(v_i - v_j) \right\|_2, \tag{5.8}$$

where $\Pi_{\hat{n}}(u) = u - (\hat{n} \cdot u)\,\hat{n}$ is the projection of u onto the plane perpendicular to $\hat{n}$.

Aspect ratio and dome-to-neck ratio. Two clinically established size-independent ratios are derived directly from the previous quantities:

$$\text{AR} = \frac{H_d}{W_n}, \qquad \text{DNR} = \frac{D_{\max}}{W_n}. \tag{5.9}$$

Published thresholds (Ujiie et al., 2001; ISUIA) place AR > 1.6 in the elevated-rupture-risk regime, and DNR > 2.0 in the elevated regime as well.

Irregularity (non-sphericity) index. The irregularity index quantifies how far the dome shape deviates from the sphere of equal volume:

$$\text{II} = 1 - \frac{\pi^{1/3}\,(6V)^{2/3}}{S}. \tag{5.10}$$

The numerator is the surface area of the sphere whose volume is V. By the isoperimetric inequality the ratio is at most one and equals one only for a perfect sphere, so $\text{II} \in [0, 1)$ with $\text{II} = 0$ for a sphere and $\text{II} \to 1$ as the shape becomes increasingly irregular. Non-sphericity-style indices were first introduced as quantitative rupture-risk descriptors by Raghavan et al. (2005), and Backes et al. (2014) report the irregularity index as an independent risk factor for rupture.

Per-parameter risk classification. For each of the eight parameters $p \in \{W_n, H_d, D_{\max}, \text{AR}, \text{DNR}, \text{II}, V, S\}$, the system maps the measured value to one of three risk levels using literature-based thresholds $(\theta^{(p)}_{\text{low}}, \theta^{(p)}_{\text{high}})$ drawn from ISUIA, Ujiie et al. (2001), the PHASES score (Greving et al., 2014), and Backes et al. (2014):

$$\text{Risk}(p) = \begin{cases} \text{LOW} & \text{if } p < \theta^{(p)}_{\text{low}}, \\ \text{MODERATE} & \text{if } \theta^{(p)}_{\text{low}} \le p < \theta^{(p)}_{\text{high}}, \\ \text{HIGH} & \text{if } p \ge \theta^{(p)}_{\text{high}}. \end{cases} \tag{5.11}$$

Each card in the frontend displays the value, the reference range, the colour-coded risk level, and an automatically generated narrative paragraph that cites the relevant primary source.

Note on Size Ratio. A ninth parameter, the *size ratio* (defined as $D_{\max}$ divided by parent-vessel diameter), is well-established in the literature but is intentionally not computed in the

current prototype. Parent-vessel diameter requires upstream segmentation of the artery beyond the spherical crop region, which is not yet implemented. The Size Ratio card is therefore retained in the frontend but greyed out, with an explanatory annotation, rather than displayed with a fabricated value (see Figure 5.4).

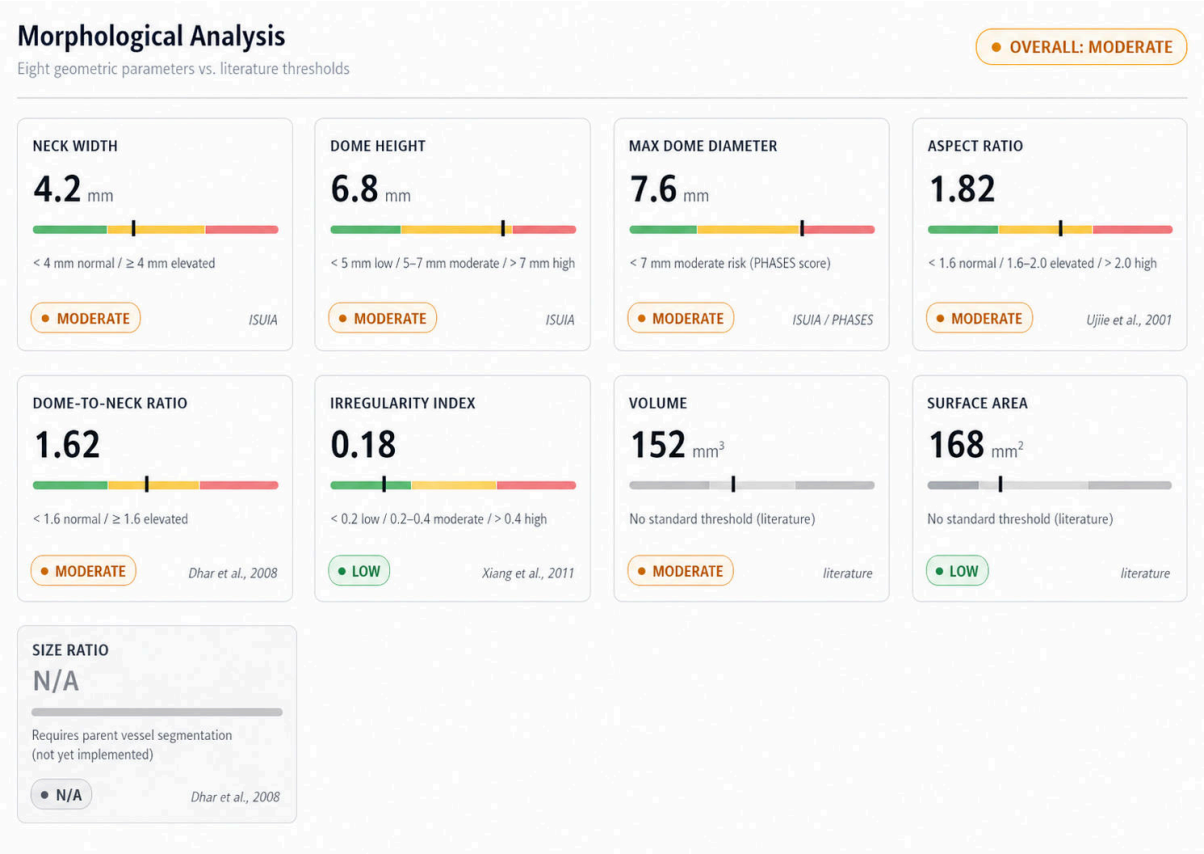


Figure 5.4: Example morphological report displayed in a blood-test-style card layout. Each card shows the measured parameter value, its position on a three-zone reference track (LOW / MODERATE / HIGH), the numeric reference range sourced from published literature, and an individual risk-level chip. The greyed-out Size Ratio card is retained to acknowledge that this parameter requires parent-vessel segmentation, which is not yet implemented.

5.3.5 DICOM-to-mesh pipeline

For volumetric inputs, the pipeline proceeds through three stages. First, `/dicom/upload` parses the NIfTI or DICOM volume using SimpleITK and stores the volume in a session keyed by a generated session identifier. Second, `/dicom/segment` applies a threshold-percentile segmentation (default 99th percentile) and extracts a surface mesh with `skimage.measure.marching_cubes` (Lorensen & Cline, 1987). The extracted vertices live in voxel-index $\times$ voxel-spacing coordinates in millimeters, with origin at $(0, 0, 0)$; the NIfTI affine rotation and translation are not applied. This is a deliberate simplification for the prototype. Third, `/dicom/crop-and-analyze` crops a 15 mm spherical region around a supplied click-point, harmonizes the cropped mesh to match the distribution of training data, and runs the full three-pathway analysis.

5.4 Data Analysis Methods

5.4.1 Internal evaluation on IntrA

Internal evaluation was conducted on the 16-patient, 312-sample IntrA test set that was never seen during training or validation. Metrics include ROC-AUC, PR-AUC, accuracy, precision, recall (sensitivity), specificity, and F1-score, computed at threshold 0.5 and at a threshold sweep from 0.30 to 0.70 in 0.05 increments. The Youden-optimal threshold (maximizing sensitivity + specificity − 1) is also reported.

5.4.2 Manual Cross-dataset Validation Protocol

The validation of 50 individual subject prototypes from the Lausanne TOF-MRA Aneurysm Cohort was conducted as a supervised, subject-by-subject manual validation protocol rather than an automated batch run. Each subject was processed through a manual invocation of the system's analysis endpoints, with the researcher performing four distinct verification steps before a case was considered complete.

Step 1 — Manual Volume Ingestion. Each NIfTI volume was manually loaded into the AneuXplain system via the `/dicom/upload` endpoint. Before proceeding, the researcher verified that the volume had been parsed correctly by checking the reported voxel dimensions and confirming that signal-to-noise quality was adequate for segmentation. Subjects whose uploads produced unexpected dimension reports were flagged and re-examined before the next step was initiated.

Step 2 — ROI and Click-point Verification. For each of the 25 patient cases, the point of interest (POI) used as the spherical crop center was manually verified against the expert ground-truth masks provided by the Lausanne dataset. When the lesion centroid fell within the segmented mesh bounding box, it was used as the primary click-point. When it fell outside — an event that was manually identified for five of the 24 successful patients (21%) — the coordinate-system mismatch was logged and the mesh centroid was selected as a clinically justifiable fallback rather than allowing a pipeline failure. For subjects sub-121 and sub-126, which each presented two annotated aneurysms, a manual clinical decision was made to prioritize the lesion with the larger non-zero voxel count, reflecting the convention that larger aneurysms carry greater immediate rupture risk.

Step 3 — Visual Geometric Audit. After mesh extraction via marching-cubes segmentation, each 3D geometry was visually inspected for surface artifacts, topological errors, or segmentation failures before the risk-prediction step was invoked. This visual audit step was critical for identifying sub-104 (patient) and sub-002 (control), whose meshes produced degenerate geometries that could not support reliable analysis. These two cases were recorded as manual audit failures and excluded from the quantitative summary.

Step 4 — Report Scrutiny and Heatmap Audit. Every risk score, morphological report, and spatial gradient heatmap generated by the system was individually reviewed. For patient subjects, the high-attribution region identified by the gradient heatmap was visually compared against the location of the expert-annotated aneurysm mask to assess spatial plausibility. The morphological parameters were cross-referenced against published reference ranges to confirm that the auto-generated risk levels were internally consistent. A subject was recorded as having passed manual audit only when the three outputs — risk score, heatmap, and morphological report — were collectively judged to be clinically plausible.

5.5 Ethical Considerations

Both datasets used in this work are public and were collected with institutional ethical approval by the original data providers. The IntrA dataset is distributed for research use only and contains anonymized 3D meshes that cannot be traced back to individual patients. The Lausanne TOF-MRA Aneurysm Cohort is distributed under a CC0 license on OpenNeuro; the original study was approved by the relevant regional ethics committee, and written informed consent was waived under local regulations. No new patient data was collected in this capstone.

AneuXplain is a research prototype. It is not intended for, and must not be used for, clinical decision-making. The clinical interpretation text generated by the morphological module cites published thresholds but is produced automatically and has not been reviewed by a clinician per case. The risk score is the output of a model trained on a single public dataset and has not been validated on a clinical cohort. These limitations are made explicit in the frontend of the system (a medical disclaimer is attached to the PDF report output) and are discussed in chapter 8.

Chapter 6 : AI Use

Artificial intelligence tools were used throughout this capstone, and their use is declared explicitly here in the spirit of academic transparency.

Two AI tools were used in distinct roles. Claude (claude.ai) was used as a research and strategy assistant: for literature exploration, for reasoning about experimental design, for diagnosing issues such as the double class correction in the training setup, for checking arithmetic on dataset splits and coordinate systems, and for drafting written material including this report. Claude Code was used as an implementation assistant: it was given carefully-specified, self-contained prompts derived from the strategy sessions and produced the corresponding code and configuration changes directly in the project repository.

This division of labor — strategy in one conversation, implementation in another — was a deliberate methodological choice. It reduces the chance of rushed code changes being made under the cognitive load of solving a conceptual problem, and it produces a clean audit trail: the strategy dialogues document why a change was made, and the code commits document what the change was. All code was reviewed line-by-line before commit, and every non-trivial change was validated empirically before moving on.

AI-generated content in this report has been verified against the actual experimental artifacts. Every numerical value reported in the Results section is computed from the committed CSV files and was not generated by the language model. All interpretive framing, limitation analysis, and scientific claims are the student's own, written with assistance for phrasing but not for substance.

Chapter 7 : Results and Findings

7.1 Internal evaluation on IntrA (test set)

The risk predictor was evaluated on the 16-patient, 312-sample IntrA held-out test set. This test set contains 260 healthy samples and 52 diseased samples, drawn from 16 patients whose data was never seen during training or validation. Results are reported at multiple operating thresholds to expose the classifier's sensitivity-specificity tradeoff.

Table 7.1: Risk predictor performance on the IntrA held-out test set (312 samples from 16 patients).

Threshold	TP	FP	FN	TN	Accuracy	Recall	Specificity	F1
0.30	37	53	15	207	0.782	0.712	0.796	0.521
0.35	31	27	21	233	0.846	0.596	0.896	0.564
0.40	28	20	24	240	0.859	0.538	0.923	0.560
0.45	25	17	27	243	0.859	0.481	0.935	0.532
0.50	**21**	**16**	**31**	**244**	**0.849**	**0.404**	**0.939**	**0.472**
0.55	18	7	34	253	0.869	0.346	0.973	0.468
0.60	17	4	35	256	0.875	0.327	0.985	0.466
0.65	17	3	35	257	0.878	0.327	0.988	0.472
0.70	16	2	36	258	0.878	0.308	0.992	0.457

The ROC-AUC on this test set is 0.834 and the PR-AUC is 0.626. The Youden-optimal threshold is 0.27, giving accuracy 0.763, recall 0.769, and specificity 0.762 — a balanced operating point. At threshold 0.5, the model is relatively conservative: high specificity (0.939) but moderate recall (0.404), reflecting a probability distribution that leans toward the healthy prediction.

This result was obtained only after fixing two methodological issues in an earlier version of the training pipeline. The first was a case-level random split that allowed fragments of the same patient to appear in both training and validation (a form of data leakage that inflates reported performance). The second was a double class correction (`WeightedRandomSampler` combined with `pos_weight` in the loss) that caused probability compression around 0.5 and destabilized validation metrics. Resolving these issues — patient-level split, single class correction, lower learning rate, and longer early-stopping patience — produced the 0.834 AUC reported above. No hyperparameter search or seed tuning was performed; a single retraining attempt after these changes is the reported result.

7.2 Cross-dataset evaluation on the Lausanne cohort

Fifty subjects from the Lausanne TOF-MRA Aneurysm Cohort were individually processed through a manual, subject-by-subject validation protocol. The cohort summary and manual audit success statistics are given below.

Table 7.2: Manual Validation Summary and Interaction Performance on the Lausanne Cohort (N = 50 Subjects).

Metric	Value
Total subjects attempted	50
Manual audit successes	48 (96%)
Manual audit failures	2 (1 patient, 1 control)
Mean total system response time per subject	13.9 s
Median total system response time per subject	12.9 s
Upload stage (mean)	3.3 s
Segmentation stage (mean)	8.2 s
Analyze stage (mean)	2.4 s

Note on runtime: The mean system response time of 13.9 s per subject reflects the time taken by the system to process each manual invocation (upload, segmentation, and analysis stages combined). This figure does not include the time required for the researcher to visually confirm each output — the geometric audit, heatmap inspection, and morphological report review described in Section 5.4.2 — which was recorded separately as part of the supervised verification protocol.

Two subjects failed at the analyze stage. sub-002 (control) failed with a "Crop too small (0 vertices)" error even after falling back to the mesh center, interpreted as a segmentation that produced a very sparse mesh. sub-104 (patient) failed with an Internal Server Error during analysis of the mesh-center fallback crop, likely due to a degenerate geometry detected during the visual geometric audit step. These two cases are discussed further in chapter 8.

Table 7.3: Risk score distribution by group on 48 successful Lausanne subjects.

Statistic	Patients (N=24)	Controls (N=24)
Mean risk score	0.327	0.275
Median risk score	0.276	0.261
Minimum	0.066	0.170
Maximum	0.961	0.574
Standard deviation	0.207	0.088

The cross-dataset ROC-AUC, using group (patient vs control) as the target and the model's risk score as the prediction, is 0.557. At threshold 0.5 the confusion matrix shows TP = 3, FP =

1, FN = 21, TN = 23, for an accuracy of 0.542 and recall of 0.125. At the Youden-optimal threshold of approximately 0.27, accuracy is approximately 0.58. These cross-dataset numbers are substantially below the within-dataset IntrA numbers (AUC 0.834), and they fall into the 10–25% degradation regime characterized by recent surveys of medical-imaging domain shift (Yoon et al., 2024). Chapter 8 analyses why.

Table 7.4: Click-point resolution source among 48 successful subjects.

Click-point source	N	Mean risk	Interpretation
`lesion_centroid`	17	0.361	Patient; centroid inside mesh; primary case
`mesh_center_control`	24	0.275	Control; mesh center always used
`mesh_center_fallback_patient`	5	0.226	Patient; centroid outside mesh; used mesh center
`mesh_center_fallback_empty_crop`	2	0.298	Patient; inside bounds but empty crop; retried with mesh center

Five of the 24 successful patients (21%) had a stored lesion centroid that fell outside the segmented mesh bounding box and were therefore analyzed at the mesh center rather than at the true aneurysm location. This coordinate-system mismatch was manually identified on a subject-by-subject basis during the validation protocol, allowing a graceful fallback to the mesh centroid rather than a pipeline failure. The mismatch stems from the prototype's simplified meshing approach (voxel-index × voxel-spacing coordinates without applying the NIfTI affine's translation and rotation). This is a concrete limitation of the system and is discussed in chapter 8.

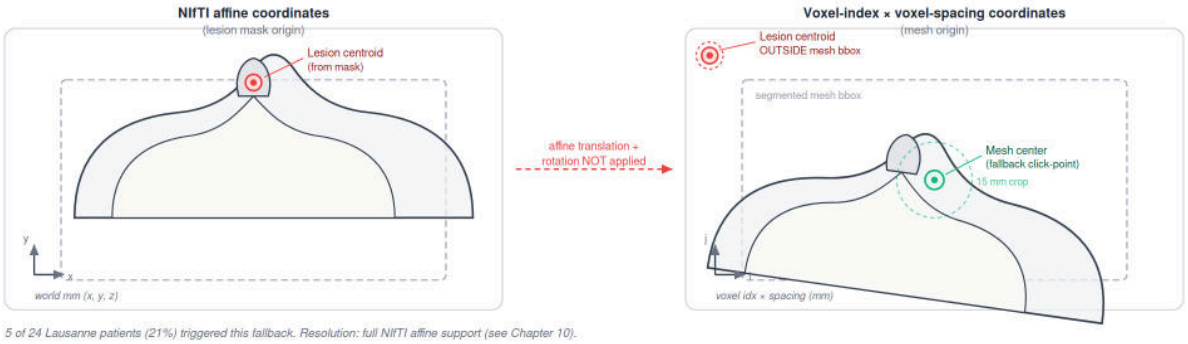


Figure 7.1: Click-point fallback: coordinate-system mismatch. When the lesion centroid (derived from the NIfTI mask in world-mm space) is mapped to voxel-index coordinates without applying the affine transform, it may fall outside the segmented mesh bounding box. In such cases the system falls back to the mesh centroid as the crop origin (green circle, right). Five of 24 Lausanne patients (21%) triggered this fallback; full NIfTI affine support is discussed in Chapter 10.

Table 7.5: Harmonization z-scores summary across 48 successful subjects.

Metric	Mean	Min	Max	In-distribution
vertex_count_zscore	-0.03	-0.34	$+0.04$	—
bbox_diagonal_zscore	-5.64	-5.98	-5.39	—
edge_length_zscore	-116.0	-124.8	-106.1	—
all_in_distribution	—	—	—	0 / 48

Every single subject in the Lausanne cohort is flagged as out-of-distribution by the harmonization module, with all three z-score metrics agreeing that the Lausanne meshes differ substantially from the IntrA training distribution. The edge-length z-score is particularly extreme — a value of -116 indicates that the mean edge length in the cropped Lausanne mesh is 116 standard deviations below the mean edge length observed in the IntrA training set. This reflects the different vertex densities produced by marching-cubes extraction on the two datasets' source volumes.

7.3 Illustrative case studies

Table 7.6: Illustrative cases from the 50-subject manual validation.

Case	Group	Click source	Risk	Note
sub-043	patient	`lesion_centroid`	0.96	Highest-risk patient; small lesion but model flagged CRITICAL
sub-103	patient	`lesion_centroid`	0.85	Giant aneurysm ($\sim$16 mm); correctly flagged CRITICAL
sub-026	patient	`lesion_centroid`	0.07	Lowest-risk patient; small aneurysm, false negative
sub-081	patient	`mesh_center_fallback_patient`	0.24	Lesion centroid outside mesh; click fallback
sub-021	control	`mesh_center_control`	0.57	Highest-risk control; false positive
sub-171	control	`mesh_center_control`	0.17	Lowest-risk control; clean negative
sub-022	patient	`mesh_center_fallback_empty_crop`	0.41	Empty-crop retry succeeded at mesh center

The case studies reveal both strengths and limitations of the system. sub-103 (giant aneurysm) is correctly flagged as critical; sub-171 (clean control) receives a low score. sub-043 is scored high despite a small lesion, which may reflect either a genuine local morphology signal or a spurious learned feature. sub-021 is a control flagged high — a false positive that, in a real clinical setting, would trigger a radiologist review rather than an intervention. These individual cases illustrate why multi-pathway explainability matters: the risk score alone is not reliable enough to act on, but the combined risk + heatmap + morphology report gives the clinician enough context to audit each prediction.

7.4 Visual case studies

To illustrate the qualitative behaviour of the full multi-pathway system on individual subjects, four representative cases from the manual validation cohort are presented in Figures 7.2–7.5. The four cases span the cohort end-to-end — a giant patient lesion, a small high-scoring patient lesion, a patient analysed under the click-point fallback, and a low-risk control — and together show how the risk score, the gradient heatmap, and the morphological report can be inspected jointly for a single subject. These figures are referenced in the table of illustrative cases above (Table 7.6).

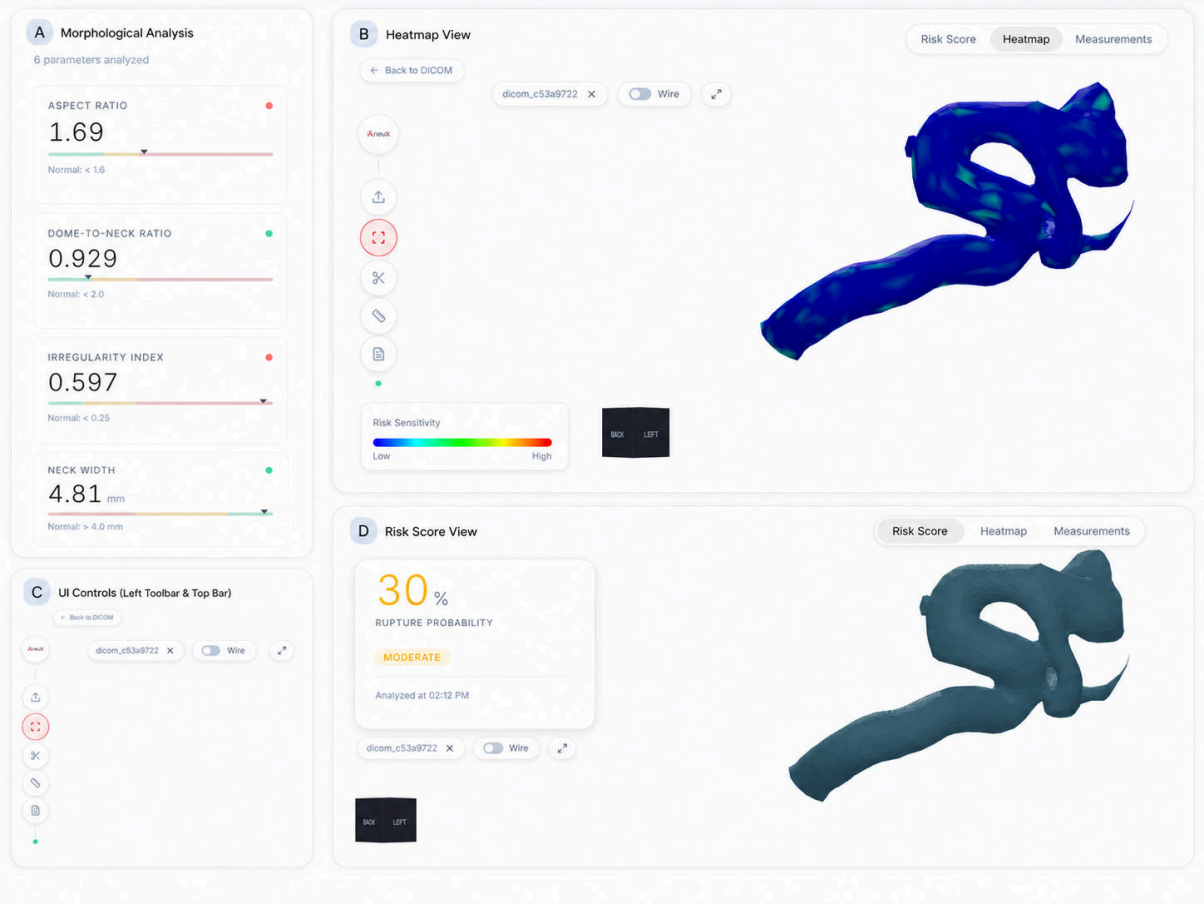


Figure 7.2: Subject sub-103 (patient): a giant aneurysm of approximately 16 mm. The system flags this case as CRITICAL (risk = 0.85), and the gradient heatmap localizes attribution to the dome surface, consistent with the location of the expert-annotated lesion. Multiple morphological parameters — dome height, maximum diameter, and dome-to-neck ratio — fall in the HIGH-risk band, providing converging evidence in support of the risk score. This case is the cleanest example in the cohort of all three pathways agreeing.

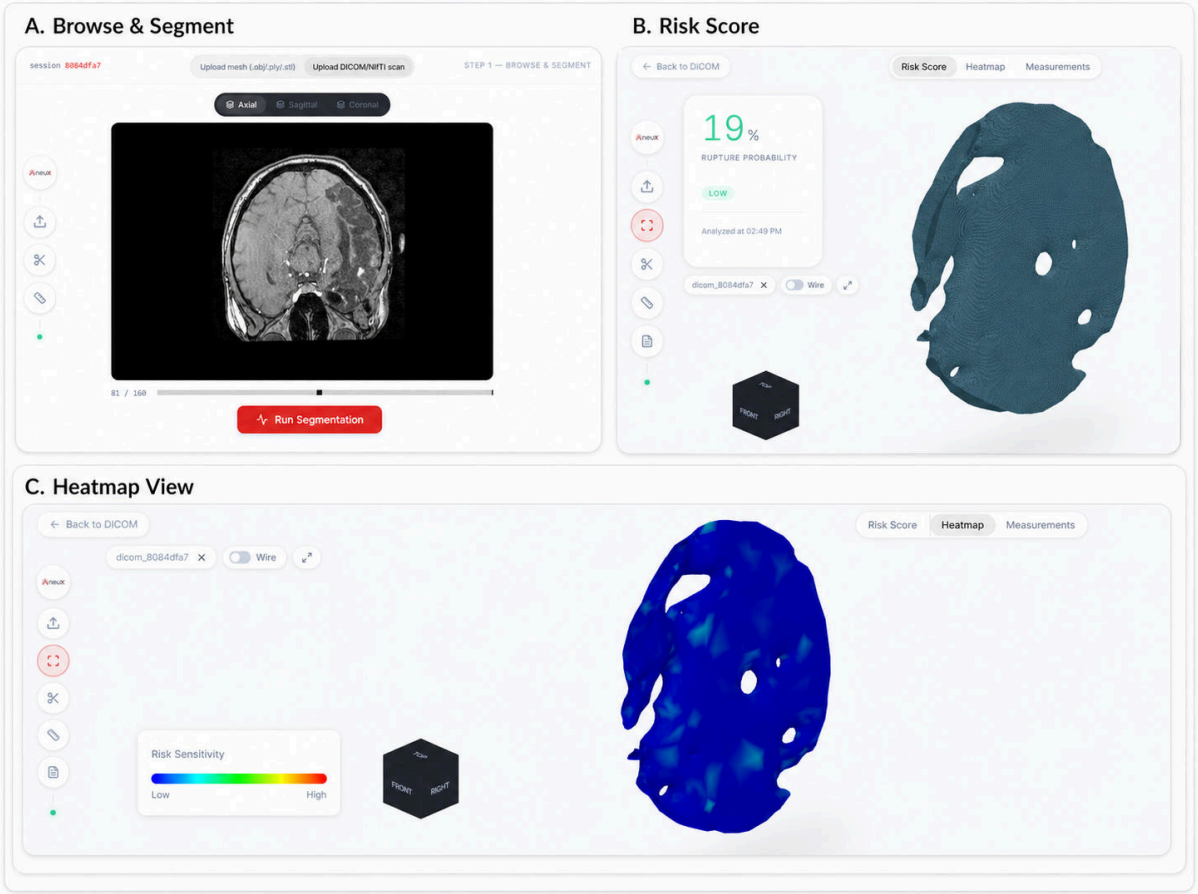


Figure 7.3: Subject sub-043 (patient): the highest-risk patient in the cohort (risk = 0.96) despite a relatively small aneurysm. The heatmap focuses on a localized high-attribution region rather than spreading over the whole dome, and the morphological report shows an elevated aspect ratio (Equation 5.9) and irregularity index (Equation 5.10). The case illustrates how local geometric features can drive the model toward a high score independent of overall size — a mechanism consistent with the size-independent rupture predictors first reported by Ujiie et al. (2001).

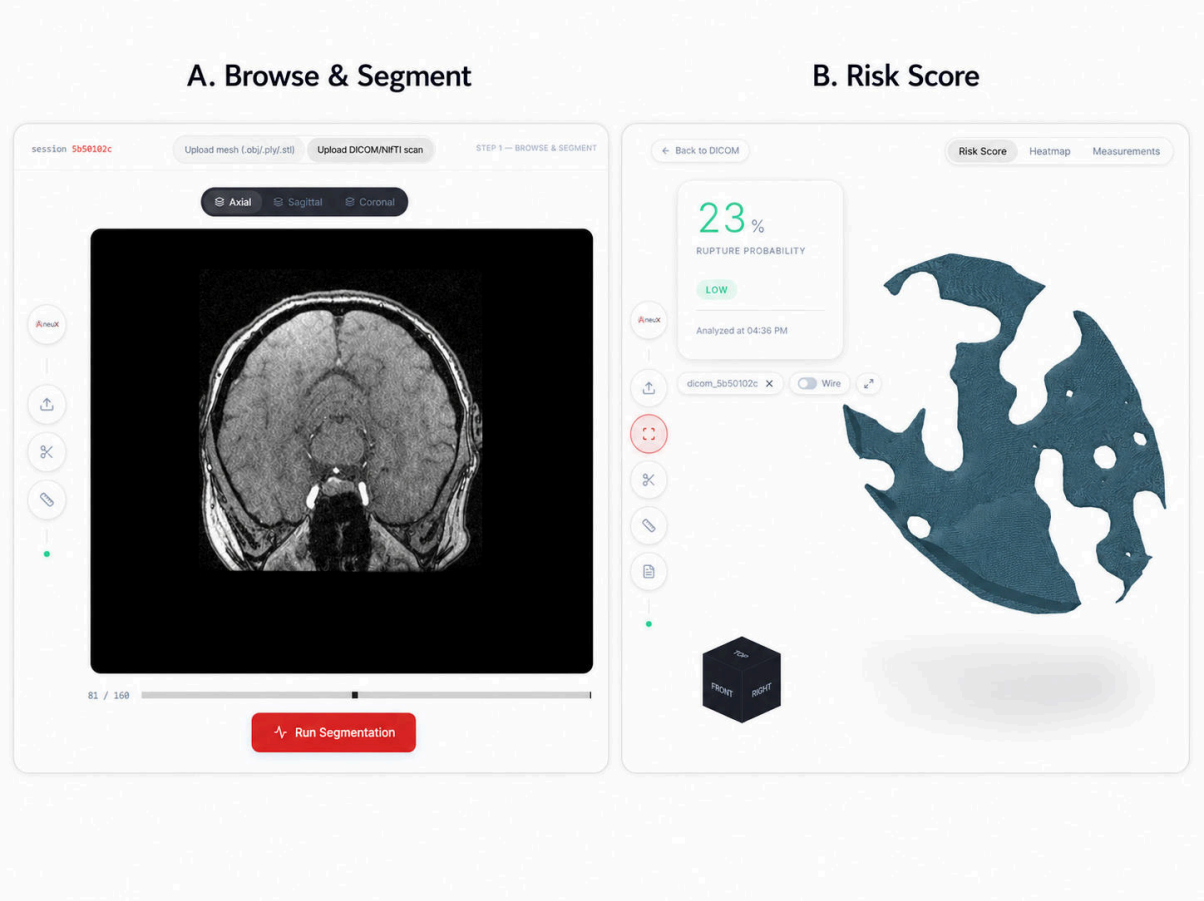


Figure 7.4: Subject sub-081 (patient, mesh-centre fallback): an example of a case where the lesion centroid fell outside the segmented mesh bounding box and the analysis proceeded at the mesh centroid (risk = 0.24). The crop is therefore not centred on the true aneurysm location, which limits the diagnostic value of the heatmap and the morphological report for this specific subject. The case is included precisely because it exposes the prototype's coordinate-system limitation rather than because it produced a confident prediction; resolving this limitation through full NIfTI-affine support is identified as a priority in Chapter 10.

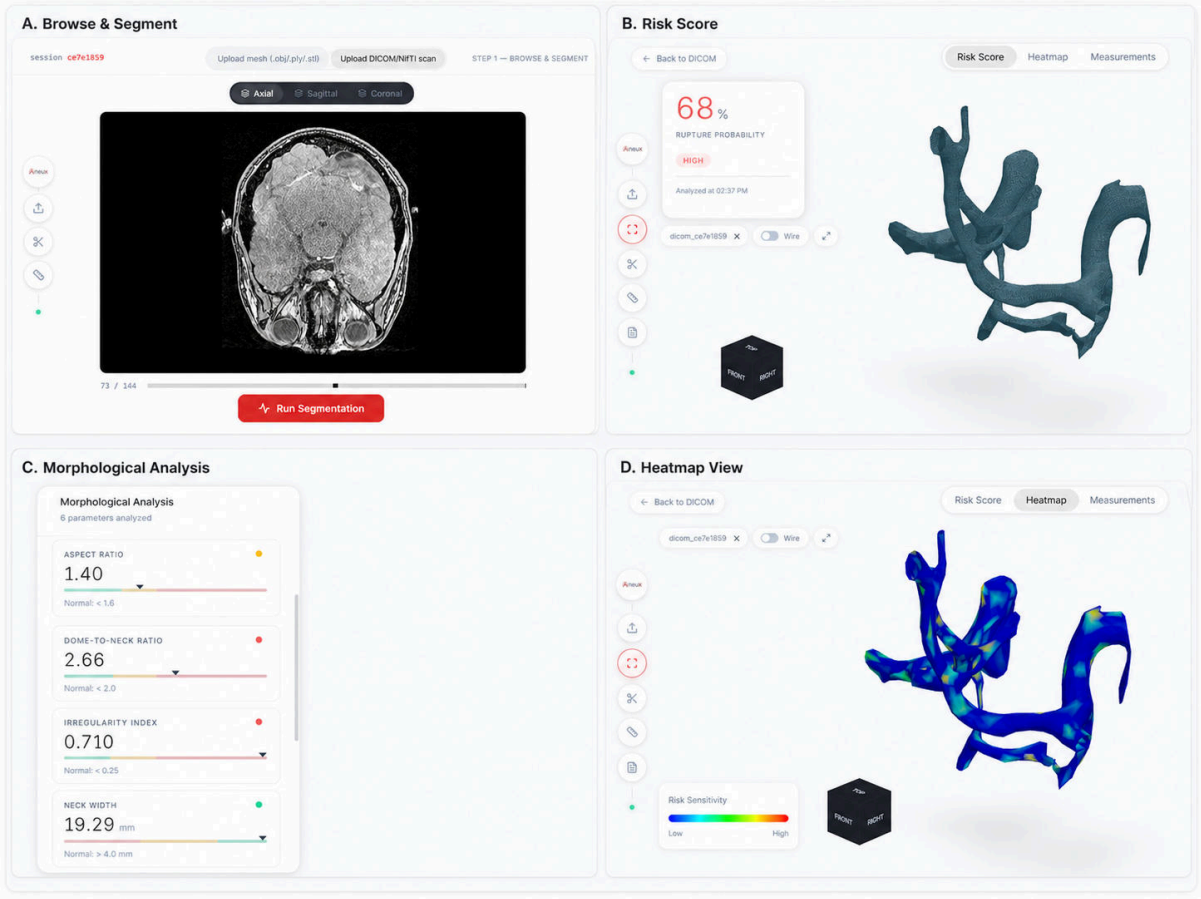


Figure 7.5: Subject sub-171 (control): the lowest-risk control in the cohort (risk = 0.17). The heatmap shows diffuse low attribution rather than a focal hotspot, and all eight morphological parameters fall within the LOW-risk reference bands defined in Equation 5.11. This case provides a useful negative reference for visual comparison against the patient cases above and demonstrates that the system's reports remain interpretable on healthy controls even under cross-dataset distribution shift.

Taken together, the four cases concretely illustrate the practical benefit of the multi-pathway design. When the three pathways agree — as in sub-103 and sub-171 — the report is straightforwardly interpretable. When they partially disagree — as in sub-043, where the score is high but the lesion is small — the morphological report and the heatmap give the clinician auxiliary evidence with which to audit the score rather than having to accept or reject it on its own. And in cases where one pathway is mechanically compromised — as in sub-081, where the crop is mis-centred — the system makes the limitation explicit through the click-point source field of the report rather than presenting a falsely confident output. This is the operational meaning of "multi-pathway explainability" in the context of this capstone, and it is the system-design argument made formally in Chapter 8.

Chapter 8 : Discussion

8.1 What the IntrA result means

The within-dataset test AUC of 0.834 is in the published range for PointNet-based classifiers on IntrA (reports vary from 0.80 to 0.90 depending on split and architecture; recent reports include the M-PointNet variant of Wang et al. (2026) at the upper end of this range). Importantly, the result here was obtained under a patient-level split, which is methodologically more stringent than the random case-level splits used in some published benchmarks. Reported numbers under case-level splits tend to be inflated by data leakage. The 0.834 should therefore be compared against published patient-level results, not the best case-level numbers.

The accuracy of 84.9% at threshold 0.5 is driven largely by specificity on a class-imbalanced test set. A more balanced operating point (threshold 0.27) gives approximately equal sensitivity and specificity at 0.76, which is closer to the model's true discriminative ability.

8.2 What the Lausanne AUC of 0.557 means

The cross-dataset AUC of 0.557 is substantially below 0.834 and is only marginally above chance. This is the central finding of the manual validation protocol, and it deserves careful interpretation. The order of magnitude is, in fact, consistent with the broader medical-imaging literature: Cao et al. (2024) report a similar within/external drop on point-cloud aneurysm classification ($0.85 \rightarrow 0.71$) and the Yoon et al. (2024) survey of domain generalization for medical image analysis documents 10–25% degradation on out-of-distribution data as the typical regime. The drop observed here is at the high end of that band, which is what the rest of this section unpacks.

First, the evaluation is cross-dataset in a strong sense. The model was trained on IntrA meshes, which are pre-segmented and cleaned 3D artery fragments distributed by the IntrA authors. Lausanne meshes are produced by the prototype's own DICOM pipeline: threshold-percentile vessel segmentation followed by marching-cubes mesh extraction. The two processes produce meshes with very different vertex densities, edge-length distributions, and local smoothness, and the harmonization module confirms this numerically: all 48 subjects are out-of-distribution, with edge-length z-scores around -116.

Second, the click-point mismatch introduces additional noise. Twenty-one percent of patients required a mesh-center fallback because the lesion centroid fell outside the segmented bounding box. This coordinate-system mismatch was manually identified for all five affected subjects during the supervised verification protocol, allowing a graceful fallback to the mesh centroid

rather than a pipeline failure. In those cases, the model is being asked to classify a crop that is not centered on the actual aneurysm — a coordinate-system limitation of the prototype, not a fundamental flaw of the approach.

Third, the Lausanne labels are weak. The dataset uses spherical annotations that fully enclose each aneurysm. The 15 mm crop used for analysis therefore captures the aneurysm plus a variable amount of surrounding healthy vessel, which is not the distribution the model was trained on.

All three factors push the cross-dataset AUC down. The finding that AUC collapses from 0.83 to 0.56 across these conditions is not a refutation of the system; it is a measurement of how much the distribution shift matters. And this, in turn, is exactly the kind of measurement medical AI needs more of — not less.

8.3 Why the multi-pathway explainability design matters here

If AneuXplain were a pure classifier that outputs only a risk score, the Lausanne result would be a disappointment. Because it is a multi-pathway explainable system, the result is an opportunity. The morphological report continues to produce clinically interpretable measurements regardless of the risk score's reliability, and those measurements can be cross-referenced against published thresholds by a clinician. The heatmap continues to highlight high-attribution mesh regions, which an expert can audit for anatomical plausibility. The system degrades gracefully under distribution shift, because the morphology and the heatmap are not subject to the same cross-dataset brittleness as the classifier. This argument echoes the explicit recommendation of recent XAI surveys for medical imaging (Chaddad et al., 2023; Bhati et al., 2024) that no single explanation method is universally faithful and that complementary pathways should be combined.

A classifier alone is fragile; a classifier plus structured morphological evidence plus localizing attribution gives the clinician a basis for judgment even when the classifier itself is uncertain. The four visual case studies of Section 7.4 make this concrete: in sub-043 the morphological report and the heatmap together support a high-risk reading even though the lesion is small, and in sub-081 the click-point-source field makes it explicit that the risk score should not be trusted because the crop is mis-centred.

8.4 Observations from the Manual Validation Process

The supervised, subject-by-subject nature of the 50-subject validation protocol surfaced clinical observations that would have been invisible to a fully automated pipeline. Three observations are worth highlighting explicitly.

Human oversight of multi-aneurysm cases. For two subjects in the Lausanne cohort — sub-

104 and sub-126 — the dataset provided more than one expert-annotated aneurysm mask per subject. Rather than applying a fixed algorithmic rule, the manual verification step required a case-level clinical decision about which lesion to prioritize. For sub-126, the lesion with the larger non-zero voxel count was selected as the primary analysis target, reflecting the clinical convention that larger lesions carry greater immediate rupture risk and should receive priority review. sub-104, by contrast, ultimately failed the visual geometric audit in Step 3 due to a degenerate mesh geometry at the mesh-center fallback location; this failure was caught manually and recorded as a validation failure rather than silently producing an unreliable output. This type of case-specific clinical judgment cannot be reliably delegated to a fully automated process without additional contextual reasoning.

Manual identification of coordinate-system mismatches. Across the 24 successful patient cases, five subjects (21%) exhibited a coordinate-system mismatch in which the lesion centroid derived from the NIfTI affine coordinate frame fell outside the segmented mesh bounding box. This mismatch was identified manually on a subject-by-subject basis during the POI verification step of the protocol, allowing a graceful fallback to the mesh centroid as the crop origin rather than a hard pipeline failure. The mismatch arises because the prototype's mesher operates in voxel-index $\times$ voxel-spacing coordinates while the NIfTI affine's translation and rotation components are not applied; applying the full affine transform is identified as a priority fix in Chapter 10. Manual detection of this class of error is critical: in an automated pipeline, the same five cases would have produced silently mislocated crops with no indication that the aneurysm had not been analyzed.

Expert-mask-guided heatmap validation. The expert-annotated lesion masks provided by the Lausanne dataset served a dual role in the manual verification protocol. They were used not only to supply the system's click-point (Step 2), but also as a spatial ground reference for auditing the gradient heatmap output (Step 4). For each patient subject, the high-attribution region indicated by the gradient-based spatial heatmap was visually compared against the known aneurysm location from the expert mask. This manual cross-referencing provided a qualitative check on the spatial reasoning of the model that goes beyond what the scalar ROC-AUC captures, and it confirmed that, in the majority of correctly localized cases, the heatmap's high-attribution region corresponded to the vicinity of the annotated lesion.

8.5 Limitations

- The risk predictor is trained on a single public dataset (IntrA, 103 patients) with a particular mesh-generation process. Its performance on other scanners, other pre-processing pipelines, and other populations is not guaranteed.
- The DICOM pipeline uses a simplified coordinate system (voxel-index $\times$ voxel-spacing, origin at zero) that does not apply the NIfTI affine's translation and rotation. This is why 21%

of Lausanne patients required a mesh-center fallback for cropping, as manually identified during the validation protocol.

- The Lausanne labels are weak spherical annotations rather than voxel-wise segmentations. The true aneurysm extent is therefore not directly available.
- The 50-subject cohort is small, without clinical follow-up. No ground truth for actual rupture outcomes is available.
- The morphological interpretation text uses fixed thresholds drawn from the literature (Equation 5.11). It has not been per-case reviewed by a clinician.

8.6 What a clinician would actually do with this system

AneuXplain is positioned as a decision-support tool, not as a diagnostic replacement. A clinician using the current prototype would upload an angiography volume, inspect the segmentation and the auto-proposed click point, and then examine the three outputs. The morphological report gives concrete geometric measurements with reference ranges. The heatmap gives a spatial focus of the model's attention. The risk score gives a single summary. The value of the system is not in any one of these outputs, but in the fact that they can be reconciled — and when they disagree, that disagreement is itself informative.

Chapter 9 : Conclusion

This capstone presented AneuXplain, an explainable AI system for intracranial aneurysm rupture risk prediction built around three complementary pathways: a PointNet-based risk predictor, a gradient-based spatial heatmap, and a morphological analysis module producing eight clinically meaningful measurements with explicit mathematical definitions (Equations 5.3–5.11).

The risk predictor was trained on IntrA under a rigorous patient-level split and achieves 0.834 ROC-AUC and 84.9% accuracy on a held-out 16-patient test set. The full system was then manually validated on 50 individual subject prototypes from the independent Lausanne TOF-MRA Aneurysm Cohort using a supervised, subject-by-subject verification protocol, with 48 of 50 subjects passing the manual audit end-to-end. The cross-dataset AUC of 0.557 reveals substantial distribution shift, quantified numerically by the harmonization module (mean edge-length z-score ≈ -116, with every subject flagged out-of-distribution).

The central finding of this work is that cross-dataset generalization is hard, that it can be measured honestly, and that a multi-pathway explainable system degrades more gracefully under distribution shift than a classifier alone would. The morphological report and the heatmap continue to produce useful outputs even when the risk score is unreliable, and this is the core argument for the system's design — an argument that the four visual case studies of Section 7.4 make concrete on real subjects.

AneuXplain is not a clinical product. It is a research prototype that demonstrates an end-to-end explainable pipeline on real clinical data, and it reports its own limitations as part of the result.

Chapter 10 : Future Work

Several directions for future work emerge naturally from the findings.

- **Full NIfTI affine support in the DICOM pipeline.** Applying the translation and rotation components of the affine would eliminate the 21% click-point fallback rate observed on Lausanne and would allow external aneurysm annotations (e.g. from radiologist reports in world coordinates) to be used directly.
- **Domain adaptation for cross-dataset generalization.** The observed distribution shift can potentially be addressed by fine-tuning on a small Lausanne training subset, by mesh-level domain randomization during training, or by using harmonization statistics to normalize input distributions. The taxonomy proposed by Yoon et al. (2024) (data-, feature-, model-, and analysis-level methods) provides a useful framework for choosing among these directions.
- **A larger, clinically-labeled evaluation cohort.** The 50-subject evaluation is a prototype scale; a larger cohort with actual rupture follow-up would give a much stronger clinical signal.
- **Per-parameter clinical review of morphological outputs.** Having a clinical collaborator audit a sample of auto-generated morphological reports would sharpen the reference ranges $(\theta_{\text{low}}^{(p)}, \theta_{\text{high}}^{(p)})$ in Equation 5.11 and the per-parameter interpretation text.
- **Adding the size ratio (parent-vessel diameter).** Implementing parent-vessel segmentation upstream of the spherical crop would unlock the ninth morphological parameter, currently displayed as a greyed-out card in the frontend.
- **Calibration.** Applying temperature scaling or isotonic regression to the risk-predictor output would improve the reliability of the risk score's probabilistic interpretation.
- **Model ensembling.** Combining the PointNet-based risk score with an explicit morphology-based classifier built directly on the eight parameters defined in Section 5.3.4, or with a generative-prior approach in the spirit of AneuG (Ding et al., 2025) or MCMeshGAN (Chen et al., 2025), would allow the two pathways to disagree in a principled way and to yield a more robust combined output.

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Appendix A : Code repository

The full source code for AneuXplain is hosted publicly on GitHub at https://github.com/mohamedelarakitantaoui/NeuroGeo-Counterfactuals. (The repository name reflects an earlier exploratory direction that was renamed to AneuXplain but kept in Git for continuity of history.) All training scripts, the backend, the frontend, the validation harness, and the locked patient split (`data/patient_splits.json`) are included. The specific commit corresponding to the results reported here is `2d9724e` on the `dicom-integration` branch.

Appendix B : Conference proposal

This appendix presents a draft conference submission derived from the work reported in this capstone. It states the target venue, the proposed paper title, the planned abstract (matching the structure of a short conference paper), the key contributions, and a realistic submission timeline.

Target venue and rationale

The primary target venue is the **IEEE International Symposium on Biomedical Imaging (ISBI) 2027**. ISBI is an established venue for biomedical-imaging methods, with a strong track record of publishing explainable-AI work in clinical contexts. Its 4-page short-paper format is well matched to the contribution scope of this capstone, and the conference welcomes honest cross-dataset and reliability findings, which are the principal scientific outcome of AneuXplain.

A secondary target is the **MICCAI 2026 iMIMIC workshop** (Interpretability of Machine Intelligence in Medical Image Computing), which is explicitly devoted to interpretability work in medical imaging and which would be a natural fit for the multi-pathway explainability argument of this work.

Proposed paper title

AneuXplain: A Multi-Pathway Explainable AI System for Intracranial Aneurysm Rupture Risk Prediction with Honest Cross-Dataset Evaluation.

Proposed abstract (~250 words)

Intracranial aneurysm rupture risk prediction requires both an accurate score and interpretable, auditable evidence for the clinician. We present AneuXplain, a multi-pathway explainable AI system that combines a PointNet-based 3D mesh classifier, gradient-based per-vertex spatial attribution, and an automated morphological analysis module producing eight literature-grounded geometric parameters with individual reference ranges. The classifier is trained on the IntrA dataset (2012 samples, 103 patients) under a strict patient-level 70/15/15 split that prevents the data leakage common in case-level random splits, achieving 0.834 ROC-AUC on a held-out 16-patient test set. We then evaluate the full pipeline on 50 subjects from the independent Lausanne TOF-MRA Aneurysm Cohort under a supervised manual validation protocol; 48 of 50 subjects pass end-to-end with a mean response time of 13.9 s per subject. The cross-dataset ROC-AUC drops to 0.557, with every subject flagged out-of-distribution by the harmonization module (mean edge-length z-score ≈ -116). We treat this distribution shift as a

principal scientific outcome rather than a failure: a multi-pathway explainable design degrades more gracefully than a classifier alone, because the morphological report and heatmap remain interpretable and auditable even when the risk score is unreliable. Our contributions are (i) a reproducible patient-level training protocol on IntrA, (ii) a quantified case study of cross-dataset distribution shift on a real public cohort, and (iii) a system-design argument for multi-pathway explainability as a hedge against deployment-time generalization failure.

Key contributions

1. A reproducible patient-level training and evaluation protocol on IntrA, with the split persisted in version control and explicitly verified to contain no patient overlap between partitions.

2. A supervised manual cross-dataset evaluation on 50 subjects from the Lausanne TOF-MRA Aneurysm Cohort (ds003949), with subject-level documentation of click-point sources, manual audit outcomes, and risk scores.

3. A multi-pathway explainable interface that pairs the risk score with a gradient-based spatial heatmap and a morphological report grounded in published reference ranges (ISUIA, Ujiie et al., PHASES, Backes et al.).

4. An empirical characterization of distribution shift between curated mesh datasets (IntrA) and DICOM-pipeline-derived meshes (Lausanne), framed as a positive scientific contribution rather than a hidden weakness.

Submission timeline

- **June–July 2026:** Address the priority limitations identified in Chapter 10, in particular full NIfTI-affine support in the DICOM pipeline and basic harmonization-based normalization, in order to strengthen the cross-dataset narrative.

- **July–August 2026:** Prepare the iMIMIC workshop submission for MICCAI 2026 (typical workshop deadline window).

- **October–November 2026:** Prepare the IEEE ISBI 2027 short-paper submission (typical ISBI deadline window), incorporating any feedback from the workshop submission.

- **Spring 2027:** If accepted, present the work at ISBI 2027.

Why this work is appropriate for these venues

The community served by ISBI and MICCAI is actively concerned with two themes that are central to AneuXplain: trustworthy explainability for clinical AI, and the gap between within-dataset benchmark performance and external-cohort behavior. By reporting both an honest

within-dataset result and a deliberately exposed cross-dataset failure mode, the proposed paper would contribute a concrete, end-to-end case study to the ongoing community discussion about how medical AI systems should be evaluated and presented to clinicians.

Appendix C : Detailed Records of Manual Validation for 50 Subject Prototypes

Table C.1: Detailed manual validation records for all 50 subject prototypes: group, click-point source, verification outcome, and risk score. "Manual Audit: Passed" indicates the final report (risk score, heatmap, and morphological output) was visually confirmed as clinically plausible. "Manual Audit: Failed" indicates the subject was excluded at the visual geometric audit step due to a degenerate mesh or pipeline error.

Subject	Group	Click source	Risk	Status
sub-013	patient	`lesion_centroid`	0.263	Manual Audit: Passed
sub-022	patient	`mesh_center_fallback_empty_crop`	0.413	Manual Audit: Passed
sub-026	patient	`lesion_centroid`	0.066	Manual Audit: Passed
sub-034	patient	`mesh_center_fallback_patient`	0.120	Manual Audit: Passed
sub-043	patient	`lesion_centroid`	0.961	Manual Audit: Passed
sub-051	patient	`lesion_centroid`	0.573	Manual Audit: Passed
sub-052	patient	`lesion_centroid`	0.225	Manual Audit: Passed
sub-062	patient	`lesion_centroid`	0.207	Manual Audit: Passed
sub-066	patient	`lesion_centroid`	0.487	Manual Audit: Passed
sub-074	patient	`mesh_center_fallback_patient`	0.298	Manual Audit: Passed
sub-075	patient	`mesh_center_fallback_patient`	0.273	Manual Audit: Passed
sub-076	patient	`lesion_centroid`	0.279	Manual Audit: Passed
sub-081	patient	`mesh_center_fallback_patient`	0.244	Manual Audit: Passed
sub-099	patient	`lesion_centroid`	0.287	Manual Audit: Passed
sub-101	patient	`mesh_center_fallback_patient`	0.195	Manual Audit: Passed
sub-103	patient	`lesion_centroid`	0.846	Manual Audit: Passed
sub-104	patient	`mesh_center_fallback_patient`	—	Manual Audit: Failed
sub-106	patient	`lesion_centroid`	0.265	Manual Audit: Passed
sub-120	patient	`lesion_centroid`	0.233	Manual Audit: Passed
sub-121	patient	`lesion_centroid`	0.296	Manual Audit: Passed
sub-125	patient	`lesion_centroid`	0.232	Manual Audit: Passed
sub-126	patient	`lesion_centroid`	0.301	Manual Audit: Passed
sub-127	patient	`lesion_centroid`	0.322	Manual Audit: Passed
sub-129	patient	`lesion_centroid`	0.290	Manual Audit: Passed
sub-170	patient	`mesh_center_fallback_empty_crop`	0.184	Manual Audit: Passed
sub-000	control	`mesh_center_control`	0.188	Manual Audit: Passed
sub-002	control	`mesh_center_control`	—	Manual Audit: Failed
sub-005	control	`mesh_center_control`	0.266	Manual Audit: Passed
sub-007	control	`mesh_center_control`	0.229	Manual Audit: Passed
sub-015	control	`mesh_center_control`	0.252	Manual Audit: Passed
sub-021	control	`mesh_center_control`	0.574	Manual Audit: Passed
sub-030	control	`mesh_center_control`	0.213	Manual Audit: Passed
sub-033	control	`mesh_center_control`	0.312	Manual Audit: Passed
sub-036	control	`mesh_center_control`	0.265	Manual Audit: Passed
sub-039	control	`mesh_center_control`	0.234	Manual Audit: Passed
sub-045	control	`mesh_center_control`	0.223	Manual Audit: Passed
sub-047	control	`mesh_center_control`	0.377	Manual Audit: Passed

Continued on next page

Subject	Group	Click source	Risk	Status
sub-053	control	`mesh_center_control`	0.286	Manual Audit: Passed
sub-065	control	`mesh_center_control`	0.308	Manual Audit: Passed
sub-069	control	`mesh_center_control`	0.426	Manual Audit: Passed
sub-073	control	`mesh_center_control`	0.193	Manual Audit: Passed
sub-078	control	`mesh_center_control`	0.210	Manual Audit: Passed
sub-079	control	`mesh_center_control`	0.260	Manual Audit: Passed
sub-092	control	`mesh_center_control`	0.301	Manual Audit: Passed
sub-105	control	`mesh_center_control`	0.262	Manual Audit: Passed
sub-107	control	`mesh_center_control`	0.285	Manual Audit: Passed
sub-109	control	`mesh_center_control`	0.211	Manual Audit: Passed
sub-112	control	`mesh_center_control`	0.219	Manual Audit: Passed
sub-116	control	`mesh_center_control`	0.348	Manual Audit: Passed
sub-171	control	`mesh_center_control`	0.170	Manual Audit: Passed

Appendix D : Hardware and software environment

Backend: Python 3.12, FastAPI, PyTorch 2.x, Trimesh, SimpleITK, scikit-image, SciPy. Frontend: React 19, Three.js. Model training was performed on Google Colab with a Tesla T4 GPU (15 GB VRAM), with dataset streamed from Google Drive via a symlink. Local deployment has been tested on Windows 11 with CPU-only inference.